2020

YEAR 11 YEARLY EXAMINATION

Mathematics Standard

Examiner: Sami El Hosri

General Instructions

- Reading Time – 10 minutes
- Working Time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A separate reference sheet is provided at the back of this paper.
- All necessary working should be shown in Questions 16 – 36

Total marks – 75

Section I: Multiple Choice
Questions 1 – 15   15 marks

- Attempt all questions
- Allow about 25 minutes for this section

Section II: Extended Response
Questions 16 – 36   60 marks

- Attempt all questions
- Allow about 1 hour 35 minutes for this section

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Section I
Questions 1 – 15 (15 marks)

Read each question and choose an answer A, B, C or D.
Allow about 25 minutes for this section.
Use the multiple-choice answer sheet for Questions 1-15.

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1. Vanessa paid \$350 to buy a dress and a pair of shoes. She paid 60% of this money for the dress. How much did she pay for the pair of shoes?

A) \$210 B) \$140 C) \$60 D) \$40

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2. At a TV factory the manager wants to use a new method of quality control. He decides to inspect every 20^{th} TV produced.

This sampling method is called:

A) Random sampling B) Self- selected sampling

C) Stratified random sampling D) Systematic sampling

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3. Kevin is a shop assistant. The table below shows some information about the hours he worked last week and his pay per hour.

Day	Hours Worked	Pay per hour	Payment
Weekdays	36	\$32	\$1 152
Saturday		\$38	
			Total \$1 418

How many hours did Kevin work on Saturday?

A) 9 B) 8 C) 7 D) 6

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4. Melissa bought a ring for \$500 and sold it later for \$525.

What was her percentage profit?

A) 5% B) 10% C) 15% D)25%

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5. Last season, a soccer team scored 30 goals in the first 9 games. After playing the remaining three games of the season the team completed their season with a mean of 4 goals per game.

How many games did the team score in the last three games?

A) 6 B)10 C) 18 D) 27

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6. The volume of a sphere is 2 100 cm^3.

What is the radius of the sphere correct to one decimal place?

A) 7.8 cm B) 7.9 cm C) 8.0 cm D) 8.1 cm

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7. Ralph measures the length of a piece of metal to be 125 cm.

What is the percentage error in his measurement?

A) ± 0.5% B) ± 0.4 % C) ± 0.04% D) ± 0.004%

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8. The difference between the longitudes of two cities P and Q is $75°$.

City P has a Coordinated Universal Time of -3.00.

Which of the following could be Coordinated Universal Time of city Q?

A) $+\ 8.00$ B) $+\ 3.00$ C) -7.00 D) $-\ 8.00$

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9. The dosage of medication for children aged 1-12 based on an adult dosage can be calculated using Young's formula shown below

$$\text{Child dosage} = \frac{\text{age of child (in years)} \times \text{adult dosage}}{\text{age of child (in years)} + 12}$$

Using this formula, the dosage of a child is 6 mL based on an adult dosage of 14 mL.

How old is this child?

A) 6 B) 7 C) 8 D) 9

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10. A basket contains 24 apples. 15 of these apples are red and 9 are green.

Pauline is to select 2 apples from the basket at random.

What is the probability that Pauline selects 2 green apples?

A) $\frac{9}{24} + \frac{8}{23}$ B) $\frac{9}{24} \times \frac{8}{23}$ C) $\frac{9}{24} + \frac{8}{24}$ D) $\frac{9}{24} \times \frac{8}{24}$

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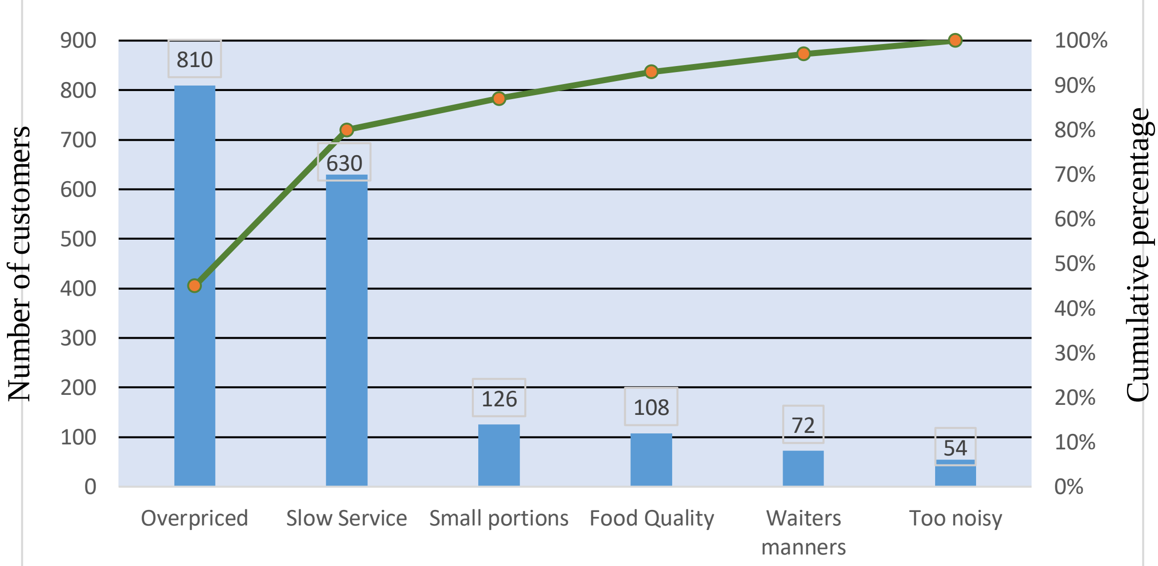
11. A restaurant owner collected data given by customers related to the reasons for being unhappy with his restaurant. The Pareto chart shows the data collected.



What percentage of the customers were unhappy because of the slow service?

A) 80% B) 45% C) 30% D) 35%

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12. A water tank containing 180 litres of water is being used to water a garden. The water is flowing out of the tank at a constant rate of 8 litres per minute.

Which of the following equations shows the volume V of the water remaining in the tank t minutes after the water started to flow out?

A) $V = -180\,t + 8$ B) $V = -8\,t + 180$

C) $V = -8\,t - 180$ D) $V = 8\,t - 180$

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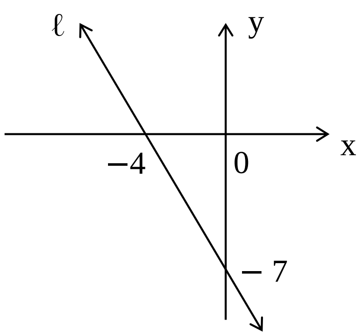
13. The graph shows a line (ℓ).

Which of the following is the equation of line (ℓ)?



A) $y = -\frac{4}{7}x - 7$ B) $y = \frac{7}{4}x - 7$ C) $y = -\frac{7}{4}x - 7$ D) $y = -\frac{7}{4}x - 4$

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14. Students in a class sat for a Mathematic test.

Because the test was considered too hard and the results were low, the teacher decided to add 6 marks to all the results.

Which of the following statements is correct?

A) The new set of marks had a larger mean, but the same standard deviation.

B) The new set of marks had a larger mean and larger standard deviation.

C) The new set of marks had the same mean and standard deviation.

D) The new set of marks had the same mean, but larger standard deviation.

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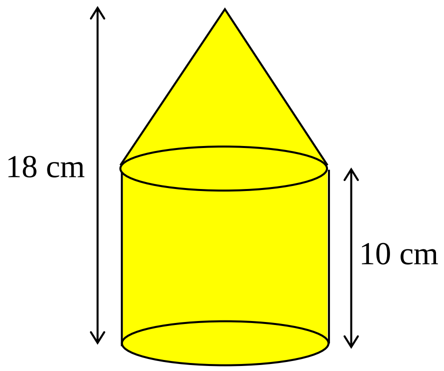
15. A solid is formed by a cone placed on the top of a cylinder.

The height of the cylinder is 10 cm and its curved area is $120\pi\ cm^2$.
The total height of the solid is 18 cm as shown in the diagram below.



What is the curved area of the cone correct to the 3 significant figures?

A) 188 cm^2 B) 251 cm^2 C) 314 cm^2 D) 565 cm^2

Year 11 MATHEMATICS STANDARD 2

Section II Answer Booklet

60 marks

Attempt Questions 16 – 36

Allow about 1 hour and 35 minutes for this section.

INSTRUCTIONS

- Answer the questions in the spaces provided. Sufficient space is provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet.

Question 16 (2 marks)

A second-hand car was purchased 3 years ago. Its value depreciates by $650 each year. **2**

The value of this car now is $6 400.

What was the purchase price?

Question 17 (2 marks)

Robert surveyed seven people. He asked them how much they paid for electricity in their last bill.

Here are the results: $750, $870, $632, $960, $642, $680, $982.

a) Calculate the mean of this set of data. **1**

b) What is the standard deviation of this set of data, correct to two decimal places? **1**

Question 18 (2 marks)

How many kilocalories (kcal) are in a can of tuna containing 5.4 grams of fat? **2**
Use the following conversions 1 gram of fat contains 37 kilojoules (kJ) and
1 kilocalorie = 4.184 kilojoules.

Question 19 (1 mark)

What is the shape of a of data distribution in which the median and the mean have values greater than the mode? **1**

Question 20 (2 marks)

City A and City B have Coordinated Universal Time of -8.00 and $+\ 4.00$ respectively. **2**

Rebecca flew from City B on a Sunday at 11:10 pm to go to City A.

The flight took 9 hours.

On what day and at what time did Rebecca's plane arrive at city A?

Question 21 (2 marks)

Michael's normal hourly rate is $28. **2**

On Sundays he is paid $60 for the first 2 hours and time and a half after that.
Last Sunday he worked 7 hours.

How much did he earn for his work on that day?

Question 22 (3 marks)

Carol wants to buy a car for \$20 000. She decides to buy it on terms as follows: **3**

- 10% deposit
- Balance and interest to be paid in equal monthly instalments of \$620 over 3 years.

What is the annual rate of simple interest that Carol will pay on the money borrowed to buy the car?

Question 23 (2 marks)

The cost C in dollars of hiring a helicopter can be calculated using a formula $C = 450 + 420\ n$, where n is the number of hiring hours. **2**

Christina paid \$800 to hire a helicopter.

For how many minutes did Christina hire the helicopter?

Question 24 (5 marks)

David starts to prepare a budget for next year.
His expected incomes are

- a fortnightly net salary of $2 460
- Simple interest earned on an investment of $20 000 at a rate of 2% per annum

His expected expenses are

- a weekly rent of $560
- a fortnightly spend for food, bills and other $360
- a weekly transport cost of $80

David's Budget Next Year			
INCOME		EXPENSES	
Salary	$63 960	Rent	B
Interest	A	Food , bills and other	C
		Transport	$4 160

a) Find the values of A, B and C. (You may assume there are exactly 52 weeks in a year.) **3**

b) David states, "even if my expenses increase by 15%, I can still save more than $15 000 next year." Is he correct ? Justify your answer using suitable calculations. **2**

Question 25 (3 marks)

Data was collected from 17 car dealers about the number of cars that they sold in the last three months. The set of data collected is displayed in the stem and leaf plot.

Cars sold in last three months

Stem	Leaf
2	2 4 8 8
3	2 4 6 6 6 8 8
4	0 0 8 8 8
6	6

a) What is the interquartile range of the data? **1**

b) Is 66 an outlier for this set of data? Justify your answer using suitable calculations. **2**

Question 26 (2 mark)

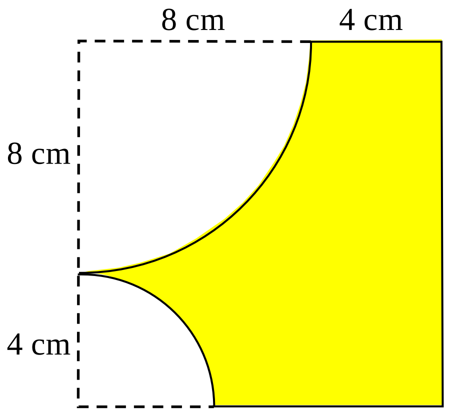
A patient received 150 mL of a medicine in one hour and 20 minutes through an intravenous drip at a constant rate. **2**

How long in hours and minutes will it take for the patient to receive a dose of 240 mL?

Question 27 (4 marks)

Kevin formed a shape by cutting two quarter-circles from a square. One of these two quarters circles has a radius 8 cm and the radius of the other is 4 cm.

The diagram shows the shape that Kevin formed.



a) Find the area of this shape, correct to two decimal places. **2**

b) Find the perimeter of this shape, correct to two decimal places. **2**

Question 28 (5 marks)

Rachel's taxable income is \$100 400. Her allowable taxable deductions amount to \$6 200. **1**

a) How much is Rachel's gross income?

b) Using the tax table below, calculate Rachel's tax on her taxable income. **2**

Taxable income	Tax on this income
0 – \$18,200	Nil
\$18,201 – \$37,000	19c for each \$1 over \$18,200
\$37,001 – \$87,000	\$3,572 plus 32.5c for each \$1 over \$37,000
\$87,001 – \$180,000	\$19,822 plus 37c for each \$1 over \$87,000
\$180,001 and over	\$54,232 plus 45c for each \$1 over \$180,000

c) Before submitting her tax return Rachel told her accountant that she forgot to claim her laptop that cost \$2 100 with her allowable taxable deductions. **2**

How much less tax on her taxable income will she pay after including the laptop in her taxable deductions?

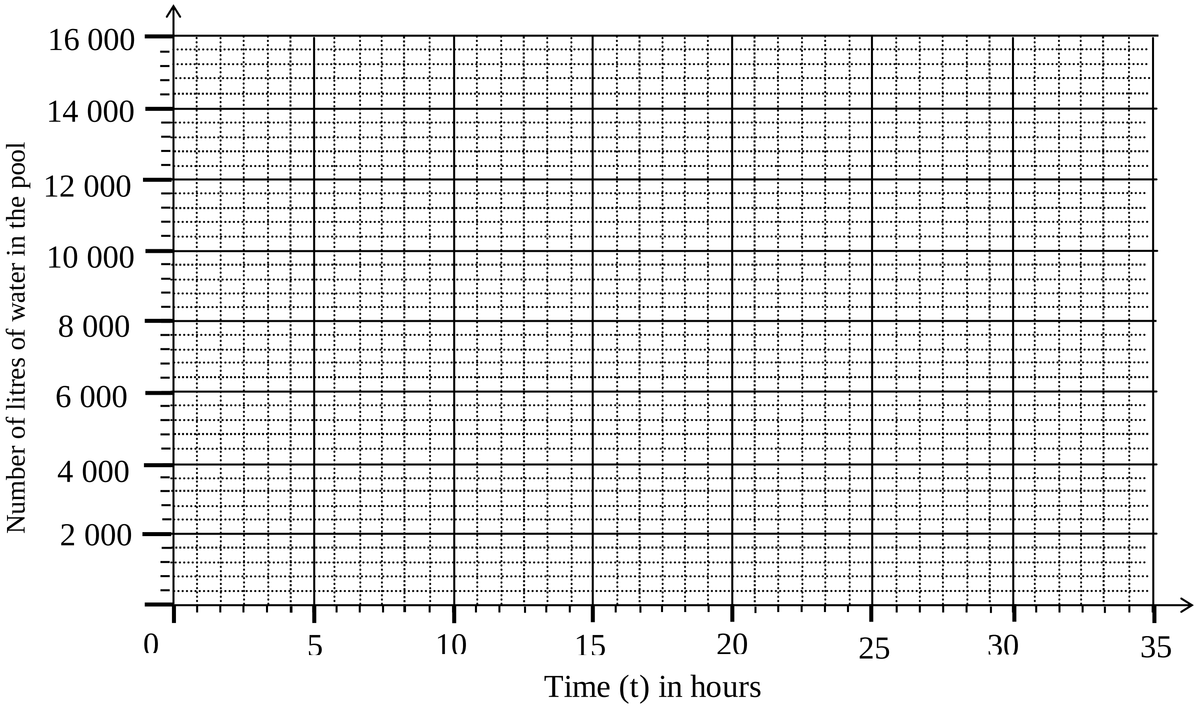
Question 29 (6 marks)

Donald has a pool in his backyard. The pool contains 16 000 L of water.
He starts to empty the pool using a pump at a constant rate of 1600 L every 5 hours.

a) Use this information to complete the table below. **1**

Time (t) in hours	0	5	10	15	20	25
Volume (V) of water in the pool in litres	16 000	14 400		11 200	9 600	

b) Plot the ordered pairs from the table above to graph the line showing the number of litres of water in the pool after t hours. **1**



c) Find the equation of the volume V of water in the pool t hours after it started to be emptied. **2**

d) After how many hours will the pool be emptied? **2**

Question 30 (2 marks)

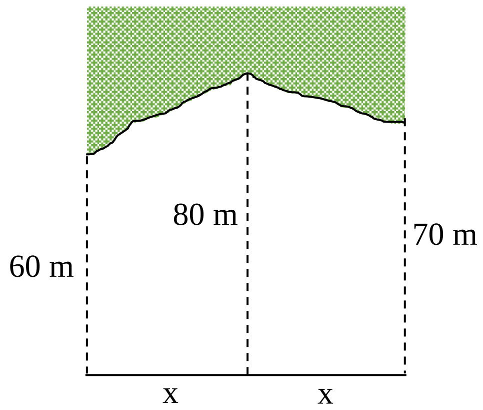
David received his holiday pay of $6 345, which is the total of 4 weeks normal pay, plus an additional 17.5% on these 4 weeks for holiday loading. **2**

What is David's normal weekly pay?

Question 31 (3 marks)

The diagram shows the land that Melissa bought near a national park. The area of this land is 6 380 m^2. **3**

By using two applications of the trapezoidal rule find the value of x.



Question 32 (4 marks)

The cost of hiring a dinner cruise for under 100 people is $C = 1\,200 + \frac{212}{5}n$,
where C is the cost in dollars and n is the number of people attending.

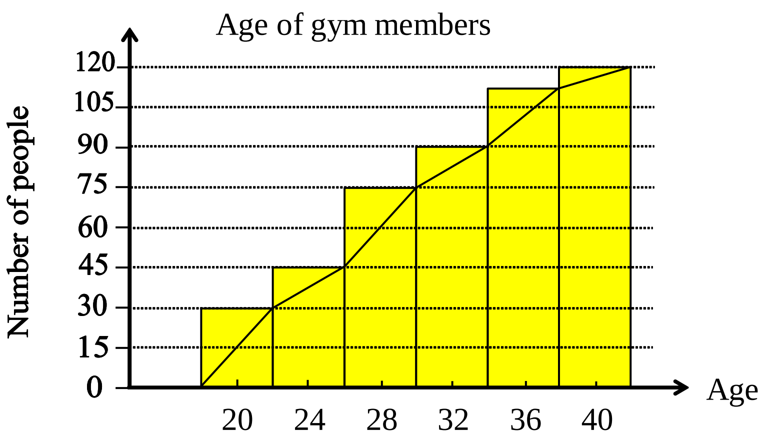
a) If the cost of a dinner cruise is $3 956, how many people attended the cruise? **2**

b) If 80 people attended the cruise and they are to share the cost, what will be the cost per person? **2**

Question 33 (2 marks)

The ages of the 120 members of a gym were recorded. These results were grouped and then displayed as a cumulative frequency histogram and polygon.

The age of the youngest member is 19 and the oldest is 42 years old.

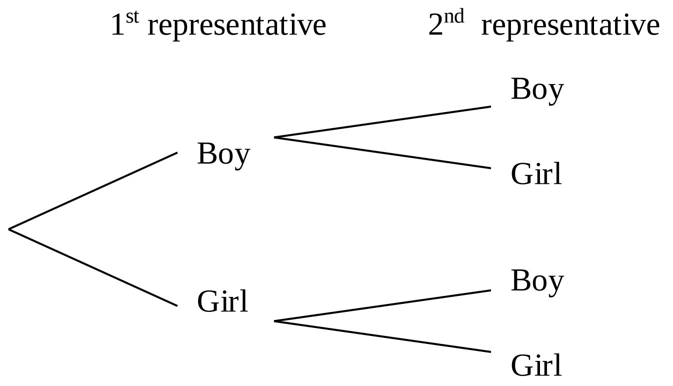


Draw a box and whisker plot to represent this data. **2**

Question 34 (2 marks)

A chess club consists of 7 boys and 5 girls. Two members are to be selected at random to represent the club in a chess competition.

a) Complete the diagram by writing the probabilities on all the branches. **1**



b) Find the probability that the two representative members are both girls. **1**

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Question 35 (3 marks)

Jared weighs 80 kg. At a party he consumed 3 bottles of a certain alcoholic beverage, each equivalent to 1.2 standard drinks. He stopped drinking at 12:30 a.m. and left the party at 2:00 a.m., when his BAC had just reached zero. **3**

Given that $BAC_{Male} = \dfrac{(10N - 7.5H)}{6.8M}$, where N is the number of standard drinks consumed, H is the number of hours of drinking and M is a person's mass in kilograms.

Also, the time needed for BAC to reach zero after the last drink consumed can be calculated using $\mathrm{T} = \dfrac{\mathrm{BAC}}{0.015}$

At what time did Jared start drinking at this party?

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Question 36 (3 marks)

The average energy used in kilojoules per kilogram of body mass by a basketball player during a leisure game is 11.98 kJ/kg per 30 minutes play. **3**

Anthony ate 2 chicken breasts each containing 162 kilocalories and drank a soft drink that contains 180 kilocalories.

Anthony weighs M kg and he needs to play a basketball game with his friends for 75 minutes in order to burn off the energy he consumed.

What is Anthony's mass M? (1 kilocalorie = 4.184 kJ)

END OF PAPER

Section II extra writing space

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REFERENCE SHEET

Measurement

Precision

Absolute error $= \frac{1}{2} \times$ precision

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

Length, area, surface area and volume

$$l = \frac{\theta}{360} \times 2\pi r$$

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(x + y)$$

$$A \approx \frac{h}{2}\left(d_f + d_l\right)$$

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$A = \frac{1}{2}ab\sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV\left(1 + r\right)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

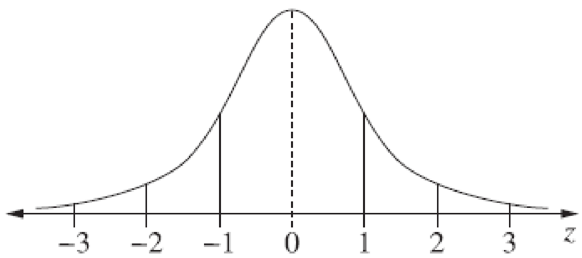
$$S = V_0\left(1 - r\right)^n$$

Statistical Analysis

$$z = \frac{x - \overline{x}}{s}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between –1 and 1
- approximately 95% of scores have z-scores between –2 and 2
- approximately 99.7% of scores have z-scores between –3 and 3